Day 2: Python Strings and the Excel Sales Check

Oct 21, 2026 | Python track + Tools track | Revision notes

Part A: Python strings

1. Joining text

- `first + " " + last` joins with a space. The space must be **inside** the quotes: `""` is empty, `" "` is one space.
- Cleanest way: an f-string, `f"{first} {last}"`. The `f` lets you put variables inside `{}`.
- **Interview edge:** strings can't be changed (immutable), so + builds a new string each time. For many pieces use `" ".join(list)`.

2. Text plus a number

- `"Age: " + age` with `age = 20` raises **TypeError**. Python will not guess between joining and adding.
- Fixes: `"Age: " + str(age)`, the f-string `f"Age: {age}"`, or a comma in print: `print("Age:", age)`.

3. String methods and immutability

Excel	Python	Example
PROPER	`.title()`	`"swagat kumar".title()` gives Swagat Kumar
UPPER / LOWER	`.upper() / .lower()`	`"abc".upper()` gives ABC
TRIM	`.strip()`	Removes spaces at both ends
LEN	`len(x)`	`len("python")` gives 6

- `name.upper()` does **not** change `name`. It returns a new string. Keep it with `name = name.upper()`.
- **Interview edge:** mutable types are list, dict, set. Immutable: str, tuple, int, float. `"they're".title()` gives `They're` (a known quirk).

4. Indexing and slicing

- Counting starts at 0. For `word = "python"`: `word[0]` is p, `word[-1]` is n. `word[6]` raises IndexError.
- Slice `word[1:4]` gives yth: the start is included, the **end is not**. Length of the slice = end minus start.
- `word[:3]` gives pyt, `word[3:]` gives hon. `word[::-1]` reverses the string.
- **Interview edge:** `word[2:100]` is fine (stops at the end), but `word[100]` is an IndexError. Python has no char type: `word[0]` is a string of length 1.

5. len and in

- `"th" in word` gives True (a bool). It is **case-sensitive**: `"TH" in "python"` is False, so use `.lower()` first.
- `word.find("th")` returns the position or -1. `word.index("th")` raises an error if the text is missing.

Python practice questions (redo from a blank file)

- Ask for a name and an age. Print one sentence with an f-string.
- Count the vowels in a typed word. It must work for uppercase letters too.
- Check if a word is a palindrome. Then make it work for a phrase with spaces.
- Take a full name like 'swagat kumar' and print the initials: S K.
- Check if an email has an @ and ends with .com, ignoring uppercase.
- Print the middle character of any string with an odd length.

Part B: Excel sales data check (Tools track)

What you built

- Cleaned 156 messy rows to 150: removed duplicate OrderIDs, fixed Category with =PROPER(TRIM(D2)), filled blank Region with Unknown, blank DiscountPct with 0.
- Revenue = Quantity x UnitPrice x (1 - DiscountPct/100). Results: total revenue 597399.2, Electronics West 39420.2, 35 orders with 15%+ discount, 70 rows met target and 80 did not. The pivot grand total matched 597399.2.

Key ideas

Topic	What to remember
Helper column + Paste Values	A formula cannot refer to its own cell (circular reference). Build the clean version in a helper column, then copy and Paste Values over the original, so deleting the helper doesn't break anything.
SUMIFS / COUNTIFS	`SUMIFS(what to add, range1, criterion1, range2, criterion2)`: the sum range comes **first**. `COUNTIFS(range, ">=15")`: operators go inside quotes.
XLOOKUP	`XLOOKUP(find this, look in this column, give back this column)`. Exact match by default. #N/A usually means a hidden space or typo in the key.
#N/A debugging	Check `LEN` vs the expected length, then `CODE(RIGHT(cell,1))`: 32 is a normal space (TRIM fixes it), 160 is a non-breaking space (use SUBSTITUTE with CHAR(160)).
Same scale	Target was stored as 18, margin as 0.18. Convert one side before comparing.
Pivot tables	The source range must stop at the last data row. A total typed under the data gets counted as a row and doubles the grand total. Always compare the pivot grand total with a plain SUM.

Interview edge: 'VLOOKUP/XLOOKUP gives #N/A but the value looks the same, why?' Answer: hidden spaces, wrong data type (text vs number), or a typo. Check TRIM, type and exact match in that order.

Excel practice questions

- Total sales per category with SUMIFS.
- Count orders above 500 in one region with COUNTIFS (two conditions).
- Look up a product price by ID with XLOOKUP, and show 'Not found' when the ID is missing (use the fourth argument).
- Clean a column of messy names with TRIM and PROPER in a helper column, then Paste Values.
- Build a pivot of sales by category and region, then check that its grand total equals SUM of the sales column.

Recall questions (answer from memory, then check)

- Why does `name.upper()` alone not change `name`?
- What does `"Age: " + 20` do, and what are 2 fixes?
- What is the difference between `word[1:4]` and `word[1:5]`?
- Why is SUMIFS written with the sum range first?
- Your XLOOKUP gives #N/A for one row. What are your first 2 checks?